# CI/CD Pipeline for Ruby on Rails Legal Application

## DevOps Project Report

Course Name: DevOps

***Institution Name:*** Medicaps University – Datagami Skill Based Course

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## Problem Statement & Objectives

### 1. Problem Statement

In modern legal-tech applications, ensuring fast, secure, and reliable deployment is critical. Traditional deployment approaches are manual, error-prone, and time-consuming, often resulting in downtime and inconsistent environments.

There is a need for an automated CI/CD pipeline that ensures:

- Continuous integration of code changes
- Automated testing for quality assurance
- Seamless deployment with minimal downtime
- Scalable and containerized infrastructure

### 2. Project Objectives

- To design and implement a CI/CD pipeline for a Ruby on Rails legal application
- To automate build, test, and deployment processes
- To ensure high code quality using automated testing
- To containerize the application using Docker
- To deploy the application on Kubernetes for scalability
- To enable continuous delivery with minimal human intervention

### 3. Scope of the Project

This project focuses on applying DevOps practices to a legal application built using Ruby on Rails.

The scope includes:

- Version control using GitHub
- CI/CD pipeline setup using GitHub Actions or Jenkins
- Containerization using Docker
- Deployment using Kubernetes
- Automated testing and build processes

## Proposed Solution

### 1. Key Features

The system provides the following features:

- **Continuous Integration (CI):** Automatically runs tests on every commit
- **Continuous Deployment (CD):** Deploys application after successful build
- **Containerization:** Uses Docker for consistent environments
- **Automated Testing:** Ensures code quality and reliability
- **Scalability:** Kubernetes handles scaling of application
- **High Availability:** Multiple pods ensure uninterrupted service
- **Automation:** Reduces manual effort and human errors
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### 2. Overall Architecture / Workflow

The workflow of the system is as follows:

1. Developer writes code and pushes it to GitHub repository
2. CI/CD pipeline is triggered automatically (GitHub Actions / Jenkins)
3. The pipeline performs:
    - Install dependencies
    - Run unit tests (Rails test suite)
4. If tests pass:
    - Docker image is built
    - Image is pushed to Docker Hub
5. Kubernetes pulls the latest Docker image
6. Application is deployed as pods in Kubernetes cluster
7. Service exposes the application to end users

### 3.Tools & Technologies Used

This project uses modern DevOps tools to automate development, testing, and deployment of a Ruby on Rails legal application.

- **Ruby&RubyonRails**
  Used to develop the legal web application with MVC architecture.

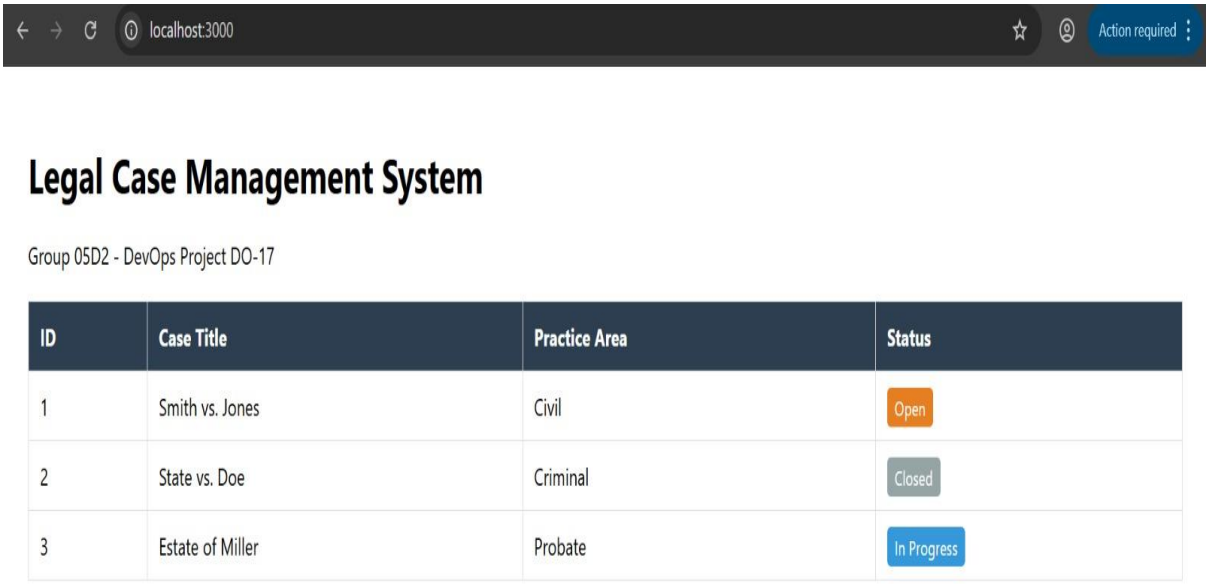
- **Git&GitHub**
  Used for version control, code management, and collaboration.
- **GitHub**
  Used to automate CI/CD pipeline including build, test, and deployment.

- **Docker**
  Used to containerize the application for consistent and portable environments. Stores Docker images for deployment.
- **Kubernetes**
  Manages deployment, scaling, and availability of the application.

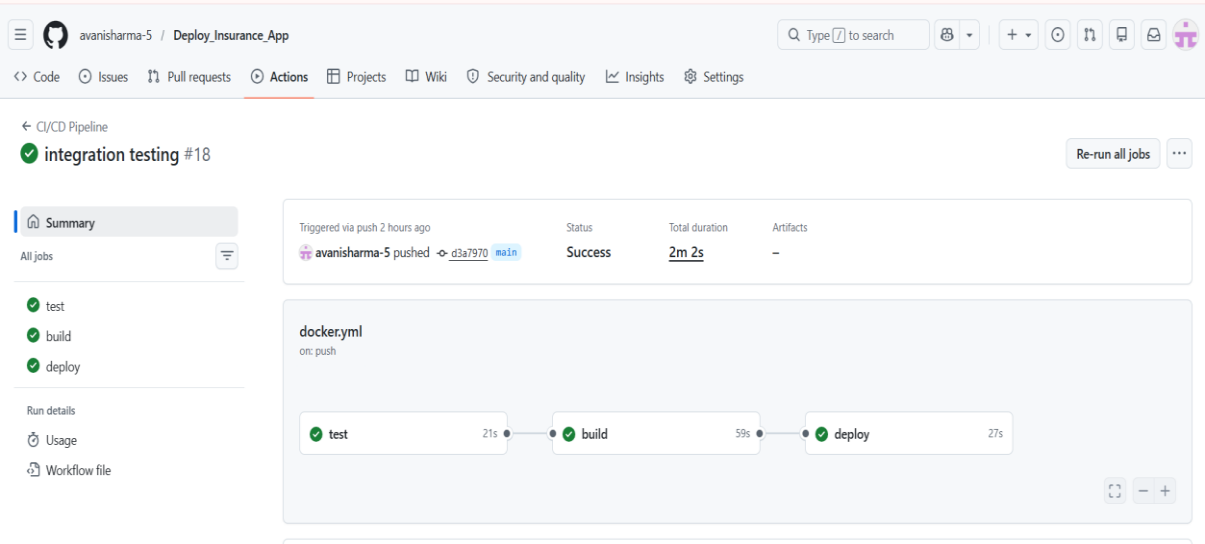
## Results & Output

**1.**Screenshots / Outputs

Deployed Application:



**Workflow:**




## Key Outcomes

- Automated testing improved code quality
- Faster deployment cycles
- Reduced manual errors
- Achieved zero downtime deployment using Kubernetes
- Scalable and reliable system architecture

# Expected Outputs

- Successful CI/CD pipeline execution
- Docker images stored in registry
- Application deployed on Kubernetes cluster
- Accessible legal application via browser

## Conclusion

This project demonstrates the implementation of a complete CI/CD pipeline for a Ruby on Rails legal application using modern DevOps tools.

By integrating GitHub, Docker, CI/CD tools, and Kubernetes:

- The development lifecycle becomes automated and efficient

- Deployment becomes faster and more reliable
- Application availability and scalability are significantly improved

Overall, the project highlights the importance of DevOps in delivering high-quality software in enterprise environments.

## Future Scope & Enhancements

- Deploy application on cloud platforms (AWS, Azure, GCP)
- Implement auto-scaling using Kubernetes HPA
- Integrate monitoring tools like Prometheus and Grafana
- Add security practices (DevSecOps)
- Implement advanced CI/CD features (blue-green deployment, canary releases)
- Add centralized logging and alerting systems